Failure of local controllability at time 21/10 for a shallow-water tank

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Abstract

We consider a one-dimensional tank governed by the normalized Saint–Venant equations, with the tank acceleration as its only control. We show that local exact controllability around the flat equilibrium, with uniformly small classical trajectories in C^1, fails at time $T = 21/10$. The terminal states form an explicit family tending to equilibrium, and the tank starts and ends with zero velocity and displacement. Thus the sufficiency of every time $T > 2$ is false. The proof computes the full quadratic endpoint in a spatial primitive, separates it uniformly from a prescribed terminal profile, and controls the nonlinear error in the same supremum norm. The least time for local controllability is not determined here.

1 The problem and the counterexample

The motion of a shallow inviscid fluid in a horizontally accelerated tank is described by the Saint–Venant equations. After normalizing the tank length, gravity and equilibrium depth to one, the system is

$$H_t + (Hv)_x = 0, \qquad v_t + (H + v^2/2)_x = -a(t), \qquad 0 < x < 1, \tag{1}$$

with

$$v(t,0) = v(t,1) = 0, \qquad s'(t) = a(t), \qquad D'(t) = s(t). \tag{2}$$

Here H is the fluid depth, v its velocity relative to the tank, and s, D are the tank velocity and displacement. The control a is continuous, and the fluid trajectory is jointly C^1 in (t, x). The mathematical question concerns classical solutions of this reduced model.

Coron proved local controllability for a sufficiently large time by the return method [1, Theorem 2]. In his formulation of the time problem [2, Problem 7.1, pp. 247–250], the conjectured range was $T > 2$. Coron, Koenig and Nguyen subsequently proved a nonlinear obstruction for $1 < T < 2$: an initially flat fluid cannot return to the flat state with the same zero tank velocity under a sufficiently small nonzero continuous control [3, Theorem 1.2]. The corresponding sufficiency question above two is asked in AIM Problem 387 [4].

We use

$$\|w\|_1 = \max_{x\in[0,1]} (|w(x)| + |w'(x)|)$$

and the state space

$$\mathcal{Y} = \left\{ (H, v, s, D) \in C^1([0,1])^2 \times \mathbb{R}^2 : H > 0,\ \int_0^1 H = 1,\ v(0) = v(1) = 0,\ H_x(0) = H_x(1) \right\}.$$

Let $P(T)$ mean that for every $\varepsilon > 0$ there exists $\eta > 0$ such that any two states whose total distance from $(1,0,0,0)$ is less than η can be joined in time T by a classical trajectory satisfying

$$\sup_{0\le t\le T} \left(\|H(t) - 1\|_1 + \|v(t)\|_1 + |s(t)| + |D(t)| + |a(t)| \right) < \varepsilon. \tag{3}$$

The conclusion below also holds with the equivalent norm $\max|w| + \max|w'|$.

Fix throughout

$$T = \frac{21}{10}, \qquad d = \frac{1}{10}, \qquad L = T - 1 = \frac{11}{10}.$$

Define $S(r) = 10r^3 - 15r^4 + 6r^5$ on $[0,1]$ and the one-periodic C^2 function b by

$$b(y) = \begin{cases} 0, & 0 \le y \le d, \\ S((y-d)/d), & d \le y \le 2d, \\ 1, & 2d \le y \le 1-d, \\ 1 - S((y-(1-d))/d), & 1-d \le y \le 1. \end{cases} \tag{4}$$

In particular, $0 \le b \le 1$. For $\rho > 0$ put

$$q_\rho(x) = \frac{3}{2}\rho b'(T-x), \qquad h_\rho(x) = \frac{q_\rho(x) + q_\rho(-x)}{2}, \qquad v_\rho(x) = \frac{q_\rho(x) - q_\rho(-x)}{2}. \tag{5}$$

Theorem 1. *There exists $\rho_* > 0$ such that, for every $0 < \rho < \rho_*$, the state $(1 + h_\rho, v_\rho, 0, 0)$ belongs to $\mathcal{Y}$, and no classical trajectory from $(1,0,0,0)$ to this state in time $T = 21/10$ satisfies (3) with $\varepsilon = 10^{-8}$. Consequently $P(21/10)$ is false, and $P(T)$ does not hold for every $T > 2$.*

The endpoints tend to equilibrium in the prescribed norm. The theorem therefore disproves local controllability, rather than imposing a lower bound on the amplitude of a particular transfer. It does not give the eventual positive controllability time.

The first-order fluid response is antiperiodic under a unit translation, whereas the target in (5) is periodic. Hence its periodic component must come from the nonlinear terms. The tank's zero displacement imposes a zero-mean condition on the temporal primitive producing that component. For a time just above two, most of the quadratic spatial primitive is the square of a single continuous function. A long positive plateau and a short zero interval cannot be approximated by such a square while preserving the zero mean. We use the full spatial primitive because its supremum error can be controlled uniformly by the trajectory's C^1 smallness.

2 Unfolding and uniform nonlinear estimates

Write $h = H - 1$ and unfold the fluid variables onto the period-two circle $\mathbb{T}_2 = \mathbb{R}/(2\mathbb{Z})$:

$$z(t,x) = \begin{cases} h(t,x) + v(t,x), & 0 \le x \le 1, \\ h(t,-x) - v(t,-x), & -1 \le x \le 0. \end{cases} \tag{6}$$

The wall conditions make z continuous and spatially Lipschitz; it is C^1 on the two open halves. Reflection is understood on $\mathbb{T}_2$. Define

$$\theta(x) = \begin{cases} -1, & 0 < x < 1, \\ +1, & -1 < x < 0, \end{cases} \qquad Q(A,B) = \frac{3A^2 - 2AB - B^2}{8}, \qquad r_z(t,x) = Q(z(t,x), z(t,-x)).$$

Direct substitution in (1) gives

$$z_t + z_x = a(t)\theta(x) - \partial_x r_z. \tag{7}$$

On the positive half, $Q(z, z \circ (-\mathrm{id})) = hv + v^2/2$; on the negative half this is the reflected flux of $h - v$. At a wall both arguments of Q agree, so $r_z = Q(A, A) = 0$. Thus the flux is continuous at each junction and (7) holds also in distributions.

For a trajectory starting at equilibrium, let

$$U(t) = \int_0^t a(\tau)\, d\tau, \qquad \ell(t,x) = \int_0^t a(\tau)\theta(x + \tau - t)\, d\tau,$$

with $U(t) = 0$ for $t < 0$. The linear response ℓ is continuous and one-antiperiodic: $\ell(t, x+1) = -\ell(t,x)$. For $0 \le x \le 1$,

$$\ell(t,x) = -U(t) + 2\sum_{k\ge 0}(-1)^k U(t - x - k). \tag{8}$$

There are at most three nonzero terms for $0 \le t \le T < 3$. With $A = \|U\|_{C^0([0,T])}$ this gives $\|\ell\|_\infty \le 7A$. Throughout this section, a supremum without a displayed domain for a time-dependent fluid variable is over $[0,T] \times \mathbb{T}_2$.

Integrating (7) along straight characteristics gives

$$z(t,x) = \ell(t,x) - \int_0^t (\partial_x r_z)(\tau, x + \tau - t)\, d\tau. \tag{9}$$

There are only finitely many wall crossings; the fundamental theorem applies on each intervening interval and the values match by continuity. Introduce the endpoint primitive

$$\Gamma_z(x) = \int_0^T r_z(t, x + t - T)\, dt. \tag{10}$$

Integrating (9) in x over any interval $[\xi, \zeta]$ and using continuity of the flux at its junctions gives

$$\int_\xi^\zeta (z(T,x) - \ell(T,x))\, dx = -\Gamma_z(\zeta) + \Gamma_z(\xi).$$

The finite integrations can be interchanged: their bounded integrands are continuous away from finitely many straight wall-crossing lines. Rectangular Riemann sums agree off the cells meeting those lines, whose total area tends to zero. Since $z(T) - \ell(T)$ is continuous, Γ_z is C^1 and

$$z(T) = \ell(T) - \partial_x \Gamma_z. \tag{11}$$

Lemma 2. *If* $\sup_t(\|h(t)\|_1 + \|v(t)\|_1) \le \varepsilon$ *and* $\varepsilon \le 1/(3T)$, *then*

$$\|z\|_\infty \le 14A, \qquad \|z - \ell\|_\infty \le 21T\varepsilon A \le 45\varepsilon A. \tag{12}$$

For pairs whose components are bounded by M *and* M' *and whose componentwise difference is at most* δ,

$$|Q(A_1, A_2) - Q(B_1, B_2)| \le \tfrac{3}{4}(M + M')\delta. \tag{13}$$

Proof. On the smooth halves, $|z_x| \le \varepsilon$ and $Q_A = (3A - B)/4$, $Q_B = -(A + B)/4$, so $|\partial_x r_z| \le (3/2)\varepsilon \|z\|_\infty$. Formula (9) gives $\|z\|_\infty \le 7A + (3/2)T\varepsilon \|z\|_\infty$. Absorbing the last term proves the first estimate; substituting it back proves the second. To obtain (13), integrate the gradient of Q on the segment between the two pairs. Its componentwise sum is bounded by $3((1-\tau)M + \tau M')/2$ at segment parameter τ. □

3 The quadratic endpoint and its separation

We first compute the quadratic primitive for a linear return. For a continuous piecewise C^1 function F supported in $[0, L]$ and vanishing at its endpoints, put $U = F + F(\cdot - 1)$ and $a = F' + F'(\cdot - 1)$ in the linear formula. The derivative may have finitely many jumps; this auxiliary linear control is used only in an explicit integral. Define

$$J_F(s) = \int_{s-1}^{s+1} F(r)\,dr, \qquad g_F(s) = F(s)^2 - F(s)J_F(s), \qquad G_F(s) = \sum_{n\in\mathbb{Z}} g_F(s+n). \tag{14}$$

The sum is locally finite and G_F is continuous and one-periodic.

Proposition 3. *For the linear response generated by this F, the primitive defined as in (10) satisfies*

$$\Gamma_\ell(x) = \frac{3}{2}G_F(T-x) + \frac{1}{2}\int_0^T U(t)^2\,dt - \frac{1}{2}\int_{\mathbb{R}} F(t)^2\,dt. \tag{15}$$

No zero-mean assumption on F is needed for this identity.

Proof. Substitution in (8) telescopes to

$$\ell(t,y) = 2F(t-y) - U(t), \qquad \ell(t,-y) = U(t) - 2F(t+y-1), \qquad 0 \le y \le 1. \tag{16}$$

Since ℓ is one-antiperiodic, its quadratic flux is one-periodic. Fix $s = T - x$ and partition time into intervals $[\beta, \beta+1]$, where $\beta = s + n$ for $n \in \mathbb{Z}$. On such an interval set $y = t - \beta$, $P = F(\beta)$ and $R = F(2t - \beta - 1)$. Expanding Q using (16) gives

$$r_\ell(t, x+t-T) = \frac{3}{2}P^2 - 2PU(t) + PR + \frac{1}{2}U(t)^2 - \frac{1}{2}R^2. \tag{17}$$

Whenever $P \neq 0$, the whole time interval lies in $[0, T]$, since $0 \le \beta \le L = T - 1$. Also

$$\int_\beta^{\beta+1} U(t)\,dt = J_F(\beta), \qquad \int_\beta^{\beta+1} R\,dt = \tfrac{1}{2}J_F(\beta).$$

Thus the first three terms in (17) integrate to $3g_F(\beta)/2$. They vanish when $P = 0$.

For the last term, the argument $r = 2t - \beta - 1 = t + y - 1$ lies in $[t-1, t]$. Hence R is zero outside $[0, T]$ and its time integration on each interval can be extended to that whole interval. An affine substitution gives

$$\sum_n \int_{s+n}^{s+n+1} F(2t - s - n - 1)^2\,dt = \tfrac{1}{2}\sum_n \int_{s+n-1}^{s+n+1} F(r)^2\,dr = \int_{\mathbb{R}} F(r)^2\,dr.$$

The length-two intervals at unit spacing cover each real point twice, apart from endpoints of no effect on the integral. Only finitely many terms intersect the support nontrivially. Summing (17) now proves (15). □

The zero tank displacement will force F to have zero mean. On the long interval $[d, 1]$, this makes G_F exactly F^2. The next lemma exploits that relation without a derivative bound for F.

Lemma 4. *Let F be continuous, supported in $[0, 1+d]$, with $\int_{\mathbb{R}} F = 0$ and $M = \|F\|_\infty > 0$. For every $c \in \mathbb{R}$ and $\rho > 0$,*

$$\|G_F - c - \rho b\|_\infty > \frac{1}{100}M^2. \tag{18}$$

Proof. Set $I = [d, 1]$, $O = [0, d]$, and for $s \in O$ write $p(s) = F(s)$, $q(s) = F(s+1)$. On I the interval $[s-1, s+1]$ contains the support, so $J_F = 0$ and $G_F = F^2$. On O the zero mean gives

$$G_F(s) = p(s)^2 + q(s)^2 + p(s) \int_s^d q(r)\, dr + q(s) \int_0^s p(r)\, dr. \tag{19}$$

If $|F|$ attains M on I, then $\max G_F \geq M^2$. Otherwise the point of maximum belongs to one of the outer intervals, and each cross term in (19) is bounded in magnitude by dM^2. Thus always $\max G_F \geq (1-2d)M^2 = 4M^2/5$. If F has a zero on I, then $\min_I G_F = 0$. If it has no zero there, it has one sign; zero mean then gives $(1-d)\min_I |F| \leq 2dM$. Consequently

$$\operatorname{osc} G_F \geq (4/5 - 4/81)M^2. \tag{20}$$

Suppose (18) fails, and put $e = 1/100$. Since b takes both values zero and one, (20) implies $\rho \geq (4/5 - 4/81 - 2e)M^2 \geq 7M^2/10$. Let $E = \int_O (p^2 + q^2)$. Integrating (19) gives

$$\int_O G_F = E + 2 \int_{0 \leq s < r \leq d} p(s) q(r)\, ds\, dr.$$

The absolute value of the cross term is at most $2 \int_O |p| \int_O |q| \leq dE$ by Cauchy–Schwarz and $2\sqrt{ab} \leq a + b$. Since $b = 0$ on O,

$$(1-d)E \leq d(c + eM^2), \qquad c \geq -eM^2, \qquad \left| \int_I F \right| \leq \sqrt{2dE}. \tag{21}$$

The last inequality uses zero mean and Cauchy–Schwarz on the two outer intervals, whose total length is $2d$.

If $c \geq 2eM^2$, then $F^2 \geq c - eM^2 \geq c/2$ on I. Therefore F has one sign and $|\int_I F| \geq (1-d)\sqrt{c/2}$. But (21) also gives $E \leq 3dc/[2(1-d)]$ and $|\int_I F| \leq d\sqrt{3c/(1-d)}$. This is impossible at $d = 1/10$, since $(1-d)^2/2 > 3d^2/(1-d)$. Hence $c < 2eM^2$, so

$$E \leq \tfrac{e}{3} M^2, \qquad \left| \int_I F \right| \leq \sqrt{e/15}\, M. \tag{22}$$

On $[2d, 1-d]$, b equals one and the presumed approximation implies $F^2 \geq \rho - 2eM^2 \geq 17M^2/25 > 16M^2/25$. Thus F has one sign on this connected interval and $|F| \geq 4M/5$. Its length is $7/10$; the remaining part of I has length $1/5$ and $|F| \leq M$. It follows that

$$\left| \int_I F \right| \geq \tfrac{7}{10} \tfrac{4}{5} M - \tfrac{1}{5} M = \tfrac{9}{25} M,$$

contradicting (22). □

4 Projection to a linear return and proof of the theorem

Suppose a trajectory reaches the target (5) with the smallness bound (3). Its tank endpoints give

$$U(T) = 0, \qquad \int_0^T U(t)\, dt = 0. \tag{23}$$

We replace its linear response by one with an exact zero endpoint and the same zero tank displacement, changing only its control primitive by a small relative amount. This allows application of Lemma 4.

For $0 \le t \le T$ define $F(t) = U(t) - U(t-1) + U(t-2)$, with negative-time values zero. Since $T < 3$,

$$F(t) + F(t-1) = U(t), \qquad \|F\|_\infty \le 3A.$$

The operators $P_\pm w(x) = (w(x) \pm w(x+1))/2$ contract the supremum norm. The unfolded terminal profile is the periodic q_ρ, so applying P_- to (12) at T gives $\|\ell(T)\|_\infty \le 45\varepsilon A$. By (8) and (23), $\ell(T, x) = 2F(T - x)$ for $0 \le x \le 1$, whence

$$\|F\|_{C^0([L,T])} \le 23\varepsilon A. \tag{24}$$

Set $F_0(t) = F(t) - (t/L)F(L)$ on $[0, L]$, zero elsewhere. It is continuous and piecewise C^1, with zero values at both support endpoints, and $\|F_0 - F\|_{C^0([0,T])} \le 23\varepsilon A$. Put $U_0 = F_0 + F_0(\cdot - 1)$ and $\alpha = \int_0^L F_0$. Since $T = L + 1$, $\int_0^T U_0 = 2\alpha$. Equation (23) gives $|\alpha| \le 23T\varepsilon A < 49\varepsilon A$.

To remove this remaining mean, use $\psi(t) = 6t(L - t)/L^3$ on $[0, L]$, zero elsewhere. Its integral is one and its supremum is $3/(2L) < 3/2$. Define

$$\widehat{F} = F_0 - \alpha\psi, \qquad \widehat{U} = \widehat{F} + \widehat{F}(\cdot - 1).$$

Then $\widehat{F}$ is supported in $[0, L]$, vanishes at its endpoints and has zero mean, while

$$\left\|\widehat{F} - F\right\|_{C^0([0,T])} \le 100\varepsilon A, \qquad \left\|\widehat{U} - U\right\|_{C^0([0,T])} \le 200\varepsilon A. \tag{25}$$

For $M = \left\|\widehat{F}\right\|_\infty$ and $\varepsilon \le 1/400$, these estimates imply $A \le 2M + 200\varepsilon A$, hence $A \le 4M$.

Let $\widehat{\ell}$ be the auxiliary linear response with primitive $\widehat{U}$. It has zero terminal fluid profile by (8), and its tank velocity and displacement endpoints vanish. Formula (8) also gives $\left\|\widehat{\ell} - \ell\right\|_\infty \le 1400\varepsilon A$. Combining (12), (13) and (25) yields

$$\left\|z - \widehat{\ell}\right\|_\infty \le 1450\varepsilon A, \qquad \left\|\widehat{\ell}\right\|_\infty \le 14A,$$

and therefore

$$\left\|\Gamma_z - \Gamma_{\widehat{\ell}}\right\|_\infty \le T\tfrac{3}{4}(28A)(1450\varepsilon A) \le 1{,}100{,}000\varepsilon M^2. \tag{26}$$

The intermediate constant after using $A \le 4M$ is 1,023,120. In particular the estimate is uniform in the frequency of the original control, whose derivative has never been bounded.

Proof of Theorem 1. The derivatives of S through order two match the constant pieces in (4). Hence q_ρ is C^1, periodic, nonzero and has zero mean. Its even and odd parts satisfy $\int_0^1 h_\rho = 0$, $v_\rho(0) = v_\rho(1) = 0$ and $h_\rho'(0) = h_\rho'(1) = 0$. For all sufficiently small $\rho > 0$, $1 + h_\rho > 0$. These are exactly the required state conditions.

For a supposed small trajectory, apply P_+ to (11):

$$q_\rho = -\partial_x P_+ \Gamma_z, \qquad P_+\Gamma_z(x) = \tfrac{3}{2}\rho b(T - x) + C_z$$

for a spatial constant C_z. Proposition 3, applied to the corrected mean-zero $\widehat{F}$, gives $\Gamma_{\widehat{\ell}}(x) = 3G_{\widehat{F}}(T - x)/2 + C_\ell$, which is one-periodic. Applying the contraction P_+ to (26), absorbing the constants and rotating the spatial variable gives some $c \in \mathbb{R}$ with

$$\left\|G_{\widehat{F}} - c - \rho b\right\|_\infty \le \tfrac{2}{3}(1{,}100{,}000)\varepsilon M^2.$$

For $\varepsilon = 10^{-8}$, all earlier smallness conditions hold and the right-hand coefficient is $11/1500 < 1/100$. If $M > 0$, this contradicts Lemma 4. If $M = 0$, then $A \le 4M = 0$, and (12) implies $z = 0$, contradicting nonzero q_ρ.

Finally, $\|h_\rho\|_1 + \|v_\rho\|_1 = O(\rho)$ with a fixed constant. Every endpoint neighborhood therefore contains a member of this unreachable family, for the same fixed $T > 2$ and the same fixed $\varepsilon > 0$. This disproves the asserted universal sufficiency. □

AI use. OpenAI Codex was used for proof development, calculation checks, retrieval of the cited literature, and preparation and editing of this manuscript.

References

[1] J.-M. Coron, *Local controllability of a 1-D tank containing a fluid modeled by the shallow water equations*, ESAIM: Control, Optimisation and Calculus of Variations **8** (2002), 513–554. doi:10.1051/cocv:2002050.

[2] J.-M. Coron, *Time for local controllability of a 1-D tank containing a fluid modeled by the shallow water equations*, Problem 7.1 in V. D. Blondel and A. Megretski (eds.), *Unsolved Problems in Mathematical Systems and Control Theory*, Princeton University Press, 2004, 247–250. doi:10.1515/9781400826155.

[3] J.-M. Coron, A. Koenig and H.-M. Nguyen, *Lack of local controllability for a water-tank system when the time is not large enough*, Annales de l'Institut Henri Poincaré C, Analyse non linéaire **41** (2024), 1327–1365. doi:10.4171/AIHPC/123.

[4] AIM problem repository, *387. The sharp local maneuvering time for a shallow-water tank*, statement revision `c9929805cece044705abaaebe1cebddac1ba11a6`. Immutable problem statement.